

MATH 332 Linear Algebra: Theory and Applications

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Quiz 2

Thursday, October 1, 2026

Instructions:

- i. This quiz has **one question**, worth **20 points**.
- ii. The time allowed is **15 minutes**. Approved testing accommodations apply.
- iii. This quiz is closed book, but you may use up to **4** double-sided letter-sized sheets of notes.
- iv. No calculators or other electronic devices are allowed. Silence and put devices in your bag under your desk or at the front of the room.
- v. Work individually. Do not communicate with other students or share materials. Direct questions to the instructor.
- vi. Show all your work to justify your answer. Answers without adequate justification will not receive credit. Simplify fractions and leave other expressions in exact form.
- vii. Print your name clearly on each answer sheet.

I understand these instructions and have not relied on any help for this exam beyond what is allowed, nor provided any help to anyone else beyond what is allowed.

Score

Signature

Date

Printed Name

1. (20 points) Let $\mathbf{A}$ and $\mathbf{B}$ be real 3×3 matrices with $\det(\mathbf{A}) = -2$ and $\det(\mathbf{B}) = 3$.

Compute each of the following, showing brief reasoning, or explain why it cannot be determined from the given information:

- a. (3 points) $\det(\mathbf{AB})$

Answer: $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}) = (-2)(3) = \boxed{-6}$.

- b. (3 points) $\det(\mathbf{A}^\top)$

Answer: Transposing preserves the determinant, so $\det(\mathbf{A}^\top) = \det(\mathbf{A}) = \boxed{-2}$.

- c. (3 points) $\det(-2\mathbf{A})$

Answer: All three columns are multiplied by -2 , so

$$\det(-2\mathbf{A}) = (-2)^3 \det(\mathbf{A}) = (-8)(-2) = \boxed{16}.$$

Multiplying the determinant by -2 only once would be incorrect.

Alternatively, write $-2\mathbf{A} = (-2\mathbf{I}_3)\mathbf{A}$ and use multiplicativity:

$$\det(-2\mathbf{A}) = \det((-2\mathbf{I}_3)\mathbf{A}) = \det(-2\mathbf{I}_3) \det(\mathbf{A}) = (-8)(-2) = \boxed{16}.$$

Here $\det(-2\mathbf{I}_3) = (-2)^3 = -8$ because $-2\mathbf{I}_3$ is diagonal. More generally, the matrix factor in $a\mathbf{A}$ is $a\mathbf{I}_3$, whose determinant is a^3 , not a . Thus the product rule gives $\det(a\mathbf{A}) = a^3 \det(\mathbf{A})$.

- d. (3 points) $\det(\mathbf{B}^{-1})$

Answer: Since $\det(\mathbf{B}) = 3 \neq 0$, $\mathbf{B}$ is invertible. Taking determinants of $\mathbf{B}\mathbf{B}^{-1} = \mathbf{I}$ gives $3 \det(\mathbf{B}^{-1}) = 1$, so $\det(\mathbf{B}^{-1}) = \boxed{1/3}$.

- e. (4 points) The volume of the parallelepiped whose three edges from one vertex are the columns of $\mathbf{B}^{-1}$

Answer: The volume is $|\det(\mathbf{B}^{-1})| = \boxed{1/3}$ cubic units.

- f. (4 points) $\det(\mathbf{A} + \mathbf{B})$

Answer: No. Determinants are not additive in whole matrices; knowing only these two determinants does not determine the determinant of their sum.

For instructor verification, take $\mathbf{A} = \text{diag}(-2, 1, 1)$. With $\mathbf{B}_1 = \text{diag}(3, 1, 1)$, the sum has determinant 4. With $\mathbf{B}_2 = \text{diag}(1, 3, 1)$, the sum has determinant -8 . Both choices satisfy $\det(\mathbf{B}_j) = 3$.

Quiz 2 Scores (out of 20)

Number of Students: 19, Minimum: 6, Maximum: 20, Mean: 11.3, Median: 10

Standard Deviation: 3.6, Quartiles (Q1, Q3): (10, 13)

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1		00000001233377
0		6689